

SonarQube Integration Project Submission

(Address Book Module)

1) Project Shared

I have integrated SonarQube Community Build with my Java project:

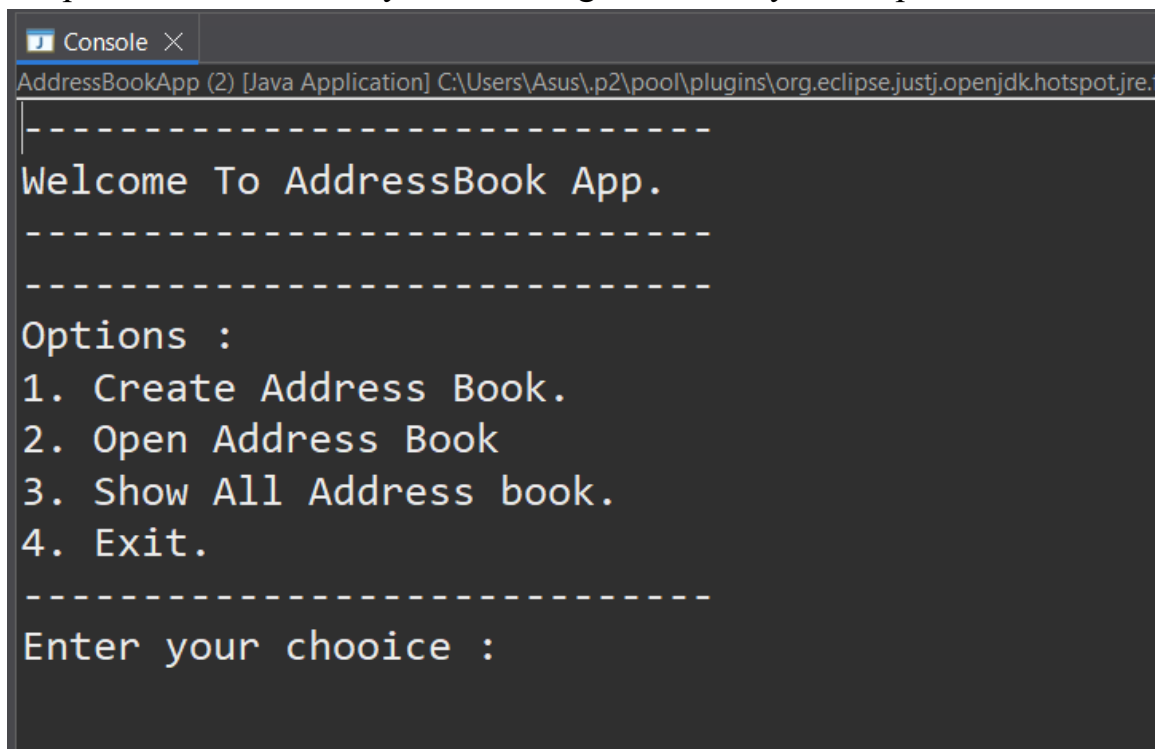
- ✓ **ProjectName:** AddressBookSystem
- ✓ **Module Included:** Address Book Module
- ✓ **Type:** Core Java + Eclipse Project
- ✓ **Source Folder:** src
- ✓ **Compiled Output Folder:** bin

2) Screenshots Provided

I have attached screenshots of:

A) Project Execution

Output of AddressBookSystem running successfully in Eclipse / Console



```
Console ×
AddressBookApp (2) [Java Application] C:\Users\Asus\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre...
-----
Welcome To AddressBook App.
-----
Options :
1. Create Address Book.
2. Open Address Book
3. Show All Address book.
4. Exit.
-----
Enter your chooice :
```

B) SonarQube Report

- SonarQube Dashboard showing:
 - Quality Gate Status: **Passed**
 - Maintainability Issues
 - Coverage and Duplication Summary

New Code

Overall Code

Security

0 Open issues

A

Reliability

0 Open issues

A

Maintainability

64 Open issues

B

Accepted issues

0

🔒

Valid issues that were not fixed

Coverage

0.0%

🔴

On 222 lines to cover.

Duplications

8.2%

🟡

On 415 lines.

Security Hotspots

0

A

C) Sonar Scanner Execution

- Command Prompt output showing:
 - **ANALYSIS SUCCESSFUL**
 - Dashboard URL
(<http://localhost:9000/dashboard?id=address-book&codeScope=overall>)

```
C:\Windows\System32\cmd.e x + ▾
23:51:30.152 INFO 6/6 source files have been analyzed for the text and secrets analysis
23:51:30.153 INFO Sensor TextAndSecretsSensor [text] (done) | time=370ms
23:51:30.161 INFO ----- Run sensors on project
23:51:30.287 INFO Sensor JavaProjectSensor [java]
23:51:30.290 INFO Sensor JavaProjectSensor [java] (done) | time=3ms
23:51:30.290 INFO Sensor Zero Coverage Sensor
23:51:30.302 INFO Sensor Zero Coverage Sensor (done) | time=12ms
23:51:30.303 INFO Sensor Java CPD Block Indexer
23:51:30.324 INFO Sensor Java CPD Block Indexer (done) | time=21ms
23:51:30.325 INFO ----- Gather SCA dependencies on project
23:51:30.332 INFO Dependency analysis skipped
23:51:30.336 INFO SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly
specify it.
23:51:30.342 INFO CPD Executor 2 files had no CPD blocks
23:51:30.343 INFO CPD Executor Calculating CPD for 4 files
23:51:30.365 INFO CPD Executor CPD calculation finished (done) | time=22ms
23:51:30.475 INFO Analysis report generated in 102ms, dir size=288.9 kB
23:51:30.507 INFO Analysis report compressed in 31ms, zip size=43.7 kB
23:51:30.528 INFO Analysis report uploaded in 21ms
23:51:30.530 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=addressbook
23:51:30.530 INFO Note that you will be able to access the updated dashboard once the server has processed the submitte
d analysis report
23:51:30.531 INFO More about the report processing at http://localhost:9000/api/ce/task?id=8f70581b-dc26-4a7b-afd9-42ff
0fac7bd2
23:51:30.549 INFO Analysis total time: 4.894 s
23:51:30.550 INFO SonarScanner Engine completed successfully
23:51:30.634 INFO EXECUTION SUCCESS
23:51:30.635 INFO Total time: 10.308s
```

3) Step-by-Step Procedure Followed (SonarQube Integration)

Step 1: Install SonarQube

1. Download SonarQube Community Build for Windows
2. Extract it to a folder (example): `C:\sonarqube-26.2.0\`
3. Open Command Prompt as Administrator
4. Go to: `sonarqube\bin\windows-x86-64\`
5. Run: `StartSonar.bat`

Step 2: Verify SonarQube is Running

1. Open browser: <http://localhost:9000>
2. Login using default credentials:
 - Username: **admin**
 - Password: **admin**
3. Set new password after login

Step 3: Create a New Project in SonarQube

1. Click **Create Project**
2. Enter Project Key: **address-book**
3. Generate a token (used for scanner authentication)

Step 4: Download SonarScanner

1. Download SonarScanner CLI for Windows
2. Extract it (example): `C:\sonar-scanner-8.0.1\`
3. Add the bin folder to system PATH: `C:\sonar-scanner-8.0.1\bin`

Step 5: Compile the Java Project

SonarQube needs .class files for Java analysis.

In Eclipse project folder, compile classes into bin.

(For Eclipse projects, bin is created automatically when project builds.)

Step 6: Run Sonar Scanner

Open Command Prompt inside the project folder:

Example: `C:\Users\...\eclipse-workspace\AddressBookSystem>`

Run:

```
sonar-scanner.bat -D"sonar.projectKey=address-book" -D"sonar.sources=src"
-D"sonar.java.binaries=bin" -D"sonar.host.url=http://localhost:9000" -
D"sonar.token=TOKEN"
```

Step 7: View the SonarQube Report

After execution, open: <http://localhost:9000/dashboard?id=address-book>

This displays:

- Code Smells
- Bugs
- Vulnerabilities
- Quality Gate Status

4) Methodology / Approach Used (Classroom Method)

Approach Followed

I followed the same approach used in the classroom:

☒ Manual SonarQube Integration using SonarScanner CLI

This method includes:

1. Running SonarQube Server locally
2. Creating project in SonarQube UI
3. Generating token
4. Running SonarScanner command from project folder
5. Viewing analysis report in SonarQube dashboard

Note about Different Approaches

Different approaches can also be used to integrate SonarQube, such as:

- SonarQube integration using **Maven**
- SonarQube integration using **Gradle**
- SonarQube integration using **Jenkins CI/CD**
- SonarLint plugin in Eclipse for live analysis

But in this project, I used the **classroom method: SonarScanner CLI**.

5) SonarQube Analysis Result Summary

From the report:

✓ Quality Gate: **PASSED**

✓ Security Issues: **0**

✓ Reliability Issues: **0**

⚠ Maintainability Issues: **64 Code Smells**

□ Coverage: **0%** (JUnit not added)

♻ Duplications: **8.2%**